PRE-ALGEBRA (7TH) · UNIT 8: LINEAR FUNCTIONS AND GRAPHING

Bike Share Showdown

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students compare bike-share and scooter rentals. Each company becomes a linear function; slope is the per-minute rate and the y-intercept is the unlock fee. Students fit ride data, solve systems by graphing, and graph a budget inequality.

The intersection of the lines tells riders exactly when the better choice switches.

Standards Addressed

- 8.F.A.1–3 Functions; linear functions
- 8.F.B.4 Construct a function to model a linear relationship; interpret slope and intercept
- 8.SP.A.1–3 Scatter plots and lines of fit
- 8.EE.C.8 Systems of equations by graphing

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with photos of city bikes and scooters. Part 1 tables, functions, and data.
Day 2	Part 2 slope, intercept, and comparing options.
Day 3	Part 3 systems and budget; write the guide.

Materials

- Student handout and Ride Data card
- Graph paper and colored pencils
- Calculators
- Optional: graphing tool

How the Three Versions Differ

Level	What changes
CP2	Tables, rules, slope and intercept meaning, and one intersection found by graphing.
CP1	Scatter plot, trend line, three options, solving systems algebraically, and a budget inequality.
Honors	Piecewise surge pricing, three-line systems, rankings, and a two-variable inequality region.

Watch For (Common Misconceptions)

- Swapping slope and y-intercept.
- Computing slope as run over rise.
- Shading the wrong side of an inequality.

Extension Ideas

- Collect real bike-share prices from a city website.
- Add a fourth company and compare.
- Graph with a free online tool and share the guide.

Answer Key – CP2

Part 1: Tables & Functions

#	Task	Answer
1	Complete a table of cost for 0, 5, 10, 15, and 20 minutes.	$1, $2, $3, $4, $5
2	Write a rule for the cost y after x minutes.	y = 0.20x + 1
3	Is cost a function of minutes? Explain.	Yes; each number of minutes gives exactly one cost.
4	Graph the CityBike rule.	Line through (0, 1), (10, 3), (20, 5).

Part 2: Slope & Intercept

#	Task	Answer
5	What is the slope of the CityBike line? What does it mean?	0.20; the cost goes up $0.20 each minute.
6	What is the y-intercept? What does it mean?	1; it costs $1 to unlock before riding.
7	ZoomScoot costs $2 to unlock and $0.15 per minute. Write its rule.	y = 0.15x + 2
8	Find the slope between (10, 3.50) and (20, 5) for ZoomScoot.	$\frac{5-3.50}{20-10} = \frac{1.50}{10} = 0.15$

Part 3: Compare

#	Task	Answer
9	Which is cheaper for a 10-minute ride? For a 30-minute ride?	10 min: CityBike $3 vs. Zoom $3.50 → CityBike. 30 min: $7 vs. $6.50 → ZoomScoot.
10	At how many minutes do they cost the same? Use your graph or a table.	20 minutes ($5 each)
11	Write two sentences for your guide telling riders which to choose.	Short rides (under 20 min): CityBike. Long rides (over 20 min): ZoomScoot.

Reflection

#	Prompt	Look for
R1	What does a steeper line mean in this situation?	The cost per minute is higher.
R2	Why do the two lines cross?	One starts higher but grows slower, so the other catches up and passes it.

Answer Key – CP1

Part 1: Functions & Data

#	Task	Answer
1	Is the set of (minutes, cost) pairs a function? Could a different set of rides make it not a function? Explain.	Yes, each minutes value has one cost. It would not be a function if the same minutes had two different costs (e.g., a surge price).
2	Make a scatter plot of the ride data. Describe the correlation.	Strong positive correlation; the points lie on a line.
3	Find the slope using two rides and write the equation of the line.	$\frac{5-3}{20-10}$ = 0.20; y = 0.20x + 1
4	Predict the cost of a 45-minute ride.	0.20(45) + 1 = $10

Part 2: Three Options

#	Task	Answer
5	Write rules for ZoomScoot and MonthPass for x total minutes in a month.	Zoom y = 0.15x + 2 per ride (assume one ride); MonthPass y = 30
6	Graph CityBike, ZoomScoot, and MonthPass (for one long ride per month) on the same axes. Where do CityBike and ZoomScoot cross?	(20, 5)
7	A rider takes 20 rides of 15 minutes each in a month. Find the monthly cost of each option.	CityBike 20(4) = $80, Zoom 20(4.25) = $85, MonthPass $30
8	Solve the system y = 0.20x + 1 and y = 0.15x + 2 algebraically to check your graph.	0.05x = 1 → x = 20, y = 5

Part 3: Budget

#	Task	Answer
9	Let r be the number of rides and m the minutes per ride (same each time). Write the cost of r 10-minute rides and an inequality for the budget.	3r ≤ 12 → r ≤ 4 rides
10	Graph y ≤ 0.20x + 1 for one ride. Which side is shaded and what does a point in the shaded region mean?	Below the line; a price you pay that is at most the CityBike cost (e.g., with a coupon).
11	Write your Rider Guide: which option is best for (a) one short trip, (b) one long trip, (c) daily commuting? Use your graphs.	(a) CityBike, (b) ZoomScoot over 20 min, (c) MonthPass.

Reflection

#	Prompt	Look for
R1	How do slope and y-intercept help you compare plans without graphing?	The intercept is the start-up cost and the slope is the rate; together they show which is cheaper short-term and long-term.
R2	Why is the intersection of two lines important here?	It is the ride length where both cost the same, the point where the better choice switches.

Answer Key – Honors

Part 1: Model Building

#	Task	Answer
1	Check every ride against CityBike's rule (y = 0.20x + 1). Do any rides not fit? What does the scatter plot look like?	All eight rides fit exactly (e.g., ride 8: 0.20(8) + 1 = 2.60), so the points lie on one line: a perfect positive linear relationship.
2	Find the slope between rides 1 and 7 and write the rule in slope-intercept form.	$\frac{7-2}{30-5}$ = 0.20; y = 0.20x + 1
3	CityBike adds a surge price: after 20 minutes, the rate becomes $0.30 per minute. Write a rule for rides longer than 20 minutes.	y = 5 + 0.30(x − 20) = 0.30x − 1
4	Is the surge-priced cost still a function? Is it still linear? Explain.	Yes, a function; it is piecewise linear (two pieces with different slopes), not a single line.

Part 2: Systems

#	Task	Answer
5	Find all ride lengths where CityBike (with surge) is cheaper than ZoomScoot.	For x ≤ 20: 0.20x + 1 < 0.15x + 2 → x < 20. For x > 20: 0.30x − 1 < 0.15x + 2 → x < 20, which is impossible. So CityBike is cheaper only for rides under 20 minutes.
6	A new company, GoLime, charges $0 to unlock and $0.25 per minute. Find where GoLime equals CityBike and where it equals ZoomScoot.	CityBike: 0.25x = 0.20x + 1 → x = 20. Zoom: 0.25x = 0.15x + 2 → x = 20. All three meet at (20, 5).
7	Explain what it means that all three lines pass through (20, 5).	At exactly 20 minutes every company costs $5; for shorter rides GoLime is cheapest and for longer rides ZoomScoot is cheapest.
8	Rank the three companies for a 6-minute and a 40-minute ride.	6 min: GoLime $1.50 < CityBike $2.20 < ZoomScoot $2.90. 40 min: ZoomScoot $8 < GoLime $10 < CityBike (surge) $11.

Part 3: Budget Region

#	Task	Answer
9	Write an inequality for s short rides and l long rides within the budget and graph it.	2.50s + 6.50l ≤ 15
10	List every whole-number combination (s, l) in the region with l ≥ 1.	(0,1), (1,1), (2,1), (3,1), (0,2)
11	Write the Rider Guide with a decision chart for short, medium, and long rides, citing the systems you solved.	Answers vary: GoLime under 20 min, any at 20, ZoomScoot over 20 min.

Reflection

#	Prompt	Look for
R1	Why might a company change its rate after 20 minutes?	To keep bikes available and earn more from long rides; it changes the slope.
R2	How does the graph of a two-variable inequality show all possible plans?	Every point in the shaded region (with whole numbers) is a combination that fits the budget.